Data Sovereignty and Community Participation in Atlantic Canadian Space Infrastructure

A critical integrative review and research agenda for launch and satellite supply chains

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Abstract

Atlantic Canadian space development raises questions about who governs the information connecting launch sites, marine activity, environmental monitoring, satellite services, and regional supply chains. This paper examines how that infrastructure could support Indigenous data sovereignty alongside distinct forms of community and organizational data governance. A critical integrative review combines Indigenous governance scholarship, official legal and policy materials, and public project documents. The proposed lens treats a consequential data relationship as its unit of analysis and asks whose authority is recognized, how people and places are represented, what permissions govern use, whether participants can exercise those permissions, how benefits and burdens are distributed, and what remedies and feedback are available. Indigenous collective rights are distinguished from individual privacy, commercial confidentiality, and general participation interests. Documents concerning Spaceport Nova Scotia provide concrete institutional entry points, but do not establish community control or demonstrate failures of compliance. Four candidate applications concern launch and fisheries coordination, environmental monitoring, procurement and workforce participation, and community-directed satellite services. Options range from bilateral agreements to shared stewardship and federated exchange; none is presumed necessary before needs are established. The paper proposes an interview and document study whose governance and community roles would be agreed with the relevant authorities before recruitment. Its contribution is an explicit bridge between governance principles, operational decisions, and assessable outcomes, rather than a claim that a new platform or national launch capability will itself produce data sovereignty.

Keywords: Indigenous data sovereignty; Mi'kma'ki; Atlantic Canada; space infrastructure; participatory data governance; launch; Earth observation

1 Purpose scope and review method

The central question is: under what institutional and technical conditions could Atlantic Canadian launch and satellite supply chains recognize and support Indigenous data sovereignty and the distinct data governance claims of surrounding communities and organizations? The question concerns both information entering the space sector and information returning from it. A fisher may need a timely notice without surrendering commercially sensitive fishing locations. A First Nation may need environmental evidence while retaining authority over knowledge contributed to interpret it. A small supplier may need access to procurement opportunities without disclosing proprietary processes. The common concern is whether data relationships enable people to influence decisions affecting them; the grounds of their claims differ.

The argument is that data sovereignty can serve as the central critical lens if its meaning is specified at each relationship. A general assertion that an ecosystem is data sovereign is too coarse. Canadian control of launch infrastructure, corporate control of mission data, and Indigenous authority over data concerning territories can coexist, conflict, or remain disconnected. The relevant inquiry follows authority into collection rules, classifications, permissions, contracts, technical controls, actual use, and avenues for

challenge. It also asks whether participants possess the resources to exercise the authority formally recognized.

The geographical anchor is Nova Scotia, particularly the Canso spaceport and its surrounding marine and community relationships. Atlantic Canada provides a wider context for satellite reception, ocean information services, research, and suppliers. This is an analytical boundary, not a claim that these actors form one governed organization. The term East Coast launch and satellite ecosystem is used descriptively. Launch safety data, spacecraft telemetry, Earth observation imagery, community knowledge, employment records, and procurement information are distinguished because their purposes and governing rules differ. The paper does not assume that satellites used locally must be manufactured or launched locally, or that launch-site approval confers authority over downstream satellite data.

This is a critical integrative review with documentary policy analysis and a proposed empirical research agenda. Targeted searches were conducted on 27 October 2026 using combinations of Indigenous data sovereignty, OCAP, CARE, Mi'kmaw research ethics, Spaceport Nova Scotia, Canso environmental assessment, Canadian commercial launch, remote sensing legislation, community data governance, fisheries satellite monitoring, data trusts, and federated data spaces. Selection prioritized original scholarly publications, Indigenous organizations' own guidance, statutes, government records, and project documents. Industry announcements were used for attributed descriptions of facilities or initiatives, not independent confirmation of performance or community benefit. The references and accompanying verification record identify the materials retained and the locations supporting the claims.

The search was iterative and purposive. It was not preregistered, did not screen all potentially eligible publications, and does not support a claim of exhaustive coverage or the absence of other regional initiatives. Statutory analysis is limited to the provisions necessary to distinguish governing regimes; a particular implementation would require assessment of its actors, contracts, data, and applicable law. The reviewed statutes and approval documents establish formal requirements; strategies and announcements record stated intentions. They do not reveal every agreement, information flow, community position, or operating practice. No interviews, community consultations, access tests, compliance inspections, or outcome evaluations were conducted for this paper.

The synthesis separates four kinds of statement: documented conditions or commitments; interpretations of those materials; proposed institutional or technical arrangements; and questions requiring empirical investigation. The analytical framework is a proposed diagnostic lens, not a validated measurement instrument or a replacement for Indigenous laws and protocols. The paper's contribution is the integration and regional application of existing insights. It does not claim to originate Indigenous data sovereignty or prove that the region lacks an effective governance system.

2 What the regional evidence establishes

Nova Scotia's environmental assessment register records the Canso Spaceport Facility Project as approved with conditions, with a ministerial decision dated 4 June 2019. Its project description also retains early vehicle and launch scheduling assumptions. Those historical plans should not be read as a current launch manifest (Nova Scotia Environment, n.d.). The associated conditions require environmental plans and monitoring, compliance reporting, complaint handling, and a community liaison committee. They also provide for a Mi'kmaq communication plan at the Department's request (Nova Scotia Environment, 2019, conditions 3.3, 4.8–4.9, 9.2, 9.4, and 10.1). These are useful entry points for researching how information informs decisions. They are neither proof of implementation nor evidence that affected Nations consented.

The project's original registration document also reports a Mi'kmaq Ecological Knowledge Study undertaken by Membertou Geomatics Solutions, drawing on knowledge-holder interviews in 2017–2018 (Strum Consulting, 2018, §8, pp. 107–108). This makes questions of authority and secondary use concrete: relevant knowledge had already entered the assessment process. The record does not establish present custodianship, permissions for reuse, or authority for a new research project. Those questions require confirmation through the relevant Mi'kmaw processes.

National investment makes the distinction between infrastructure sovereignty and data authority especially consequential. On 16 March 2026, National Defence announced a ten-year, $200 million agreement to lease a dedicated launch pad at Spaceport Nova Scotia. The release required the dedicated pad and associated services to reach initial operational capability by the end of 2026 and described a separate initial light-lift capability target for 2028 (Department of National Defence, 2026). The release also required at least 90% of the funds received to be spent in Canada. These are dated announcements and targets. They do not establish that an operational orbital launch capability existed when this review was conducted. Nor does Canadian expenditure establish local access, Indigenous governance, or equitable distribution of benefits.

Transport Canada's consultation ran from 25 February to 13 April 2026 and sought feedback on draft launch application requirements, including public notification (Transport Canada, 2026b). The linked application guidance remained explicitly marked draft when reviewed (Transport Canada, n.d.). The Department's overview states that it does not license spaceports and that launches require ministerial permission (Transport Canada, 2026a). A prospective regional governance arrangement needs to track the instruments actually applying to each activity. Safety authorization, environmental approval, remote sensing licensing, and a commercial service agreement answer different questions.

The ecosystem also includes facilities beyond the launch event. Leaf Space announced completion of its first Canadian ground station at Spaceport Nova Scotia on 8 September 2026 and stated that it would support S-band and X-band data downloads for satellite operators (Leaf Space, 2026). This is a provider's account of infrastructure, not evidence of an entitlement for nearby communities to receive satellite products. It nevertheless illustrates an important analytical separation: hosting a receiving facility does not, by itself, establish who controls acquisition, processing, access, or commercialization.

Community engagement is likewise an institutional starting point rather than an outcome measure. Maritime Launch's September 2026 recruitment notice for its community liaison committee sought participation including fishers and local organizations (Maritime Launch Services, 2026). The research question is what the committee can actually change, what information it receives, whom members represent, and how their input is answered. Its existence alone cannot establish either meaningful participation or its absence.

These documents justify a study of specific governance interfaces. They do not justify a presumption that all relevant information is centralized, unavailable, mishandled, or ungoverned. Private agreements and community protocols may already address some issues. Conversely, publication of a plan or announcement cannot settle whether its recipients can understand, contest, or use the resulting information. That distinction defines the empirical work still needed.

3 Constructing a defensible data sovereignty lens

Authority must be distinguished from location and access

Hummel et al. (2021, Discussion and Table 5) show that data sovereignty has several meanings, involving different agents, values, rights, and practical abilities. Their review supports conceptual specification; it does not validate one governance architecture. Here, data sovereignty concerns the authority and practical capacity to govern data relationships. Physical storage is one possible condition of that capacity, not its definition. A locally hosted database could still be governed entirely by an external contractor. A community could retain meaningful decision rights while using a service elsewhere, subject to applicable legal and contractual constraints.

Access is also insufficient. Being allowed to download a dataset does not determine who set its categories, whether its collection was authorized, which inferences are legitimate, or whether a harmful representation can be corrected. Conversely, a refusal to share some information may be an exercise of legitimate governance. The analysis must consider data for community governance as well as governance of data: what information a Nation or organization needs to make its own decisions, not simply what it must protect from others.

Indigenous data sovereignty has a distinct foundation in collective self-determination and authority concerning peoples, territories, resources, and knowledge. UNDRIP recognizes participation through Indigenous Peoples' own representative institutions and rights concerning cultural heritage and knowledge; Articles 19 and 32 also direct States to consult and cooperate in good faith with the Indigenous Peoples concerned to obtain consent in the circumstances specified there (United Nations, 2007, arts. 18–19, 31–32). These provisions should not be reduced to a generic stakeholder consultation checklist. Their legal and institutional implementation requires attention to the relevant Peoples, decision, and jurisdiction.

The First Nations principles of OCAP® concern ownership, control, access, and possession. FNIGC explicitly identifies them as First Nations principles and emphasizes interpretation through each Nation's own worldview and protocols (First Nations Information Governance Centre [FNIGC], n.d.). They should not be presented as a universal template for all Indigenous Peoples, let alone all nearby firms or fishing associations. OCAP® is a registered trademark of the First Nations Information Governance Centre. Its invocation in a proposal does not establish endorsement or compliance.

The CARE principles add Collective Benefit, Authority to Control, Responsibility, and Ethics to discussions of data stewardship. They direct attention to people, purposes, relationships, and consequences, complementing the FAIR emphasis on making data findable, accessible, interoperable, and reusable (Carroll et al., 2020, The CARE Principles). Jennings et al. (2025) extend the relevance of Indigenous governance to open Earth systems science, including information concerning lands and environments. Their perspective is a reason to examine remote sensing relationships carefully, not empirical evidence that a particular Atlantic project has caused harm.

Shared questions do not imply equivalent rights

Other participants may have governance claims with different foundations. Individuals may hold privacy rights; workers may seek protection against intrusive monitoring and fair access to training; fishers may seek confidentiality, safety information, and influence over coordination; firms may hold contractual and intellectual property interests. Municipalities and public bodies may also have statutory duties relevant to

particular data activities. Indigenous Nations are not simply another stakeholder category within this list. A governance arrangement must identify the source and scope of each authority rather than award identical powers to every participant. The evaluation adopts self-determination, respect for applicable rights, meaningful participation, and attention to the distribution of benefits and burdens as normative commitments; the review does not derive those commitments from measured local outcomes.

Nor is a community necessarily homogeneous. Leaders, knowledge holders, harvesters, workers, businesses, and people outside formal organizations may disagree. Representation must be authorized for the question at hand. One person's consent cannot automatically authorize disclosure of collective knowledge; collective authorization does not erase an individual's interests or applicable rights. A commercial partner cannot establish its representative mandate merely by offering a seat on an advisory committee.

Three implications follow. First, the analysis should distinguish established authority, asserted authority, obligations, and participation interests, recording their basis and any dispute. Recording an authority in a sector process does not create it, and an operator's refusal to recognize it does not by itself extinguish it. Second, disputes need an accountable process rather than silent resolution by a platform administrator. Third, a shared framework should provide room for different governance arrangements. Common metadata and exchange procedures may be useful without requiring one institution to own every dataset or make every decision.

The unit of analysis is a consequential data relationship

The proposed unit is a data relationship at a decision point: for example, an operator requesting fishing activity information to plan coordination, or a research team proposing to combine satellite imagery with community observations. The relationship is traced across collection, classification, linkage, inference, access, use, onward sharing, retention, and disposal. A single dataset may participate in several relationships with different authorities and purposes. A derived map or model should be examined again rather than assumed to inherit an unrestricted right of reuse.

Table 1 translates this lens into seven diagnostic questions. The dimensions synthesize the literature and the regional problem; they are proposed analytical categories. They should be revised with participants and should not be collapsed into one sovereignty score.

Table 1 Questions for examining a data relationship

Dimension	Diagnostic question	Observable evidence
Authority	Who can authorize, refuse, or require this activity, and on what basis?	Mandates, laws, protocols, agreements, named decision makers
Representation	Who defines the categories, interpretation, and acceptable inference?	Data dictionaries, map legends, quality notes, review records
Permissions	What collection, access, reuse, linkage, retention, and disclosure are permitted?	Purpose statements, permissions, contracts, exception records
Capability	Can the relevant people understand and exercise their powers?	Paid time, staff, connectivity, accessible formats, independent advice
Consequences	Who gains, who bears cost or risk, and who defines benefit?	Distributional assessments, costs, locally selected outcome measures

Dimension	Diagnostic question	Observable evidence
Accountability	Can a decision be questioned, corrected, or remedied?	Reasons, logs, complaint routes, escalation and response records
Feedback	Can findings and challenges change the rules and practice?	Revisions, decision histories, follow-up assessments

In this formulation, being recognized by an ecosystem means more than being mentioned in a strategy. It means relevant authority is respected in decisions and affected parties can obtain an accountable response to concerns. These are different standards: a binding authority must be respected according to its scope, while a participation interest does not automatically confer a right to determine the outcome. A useful trace is: an authority or need is articulated; the responsible party records it; an authorized person determines a response; that response changes a rule or explains a refusal; and the result can be reviewed. A valid statutory obligation may limit a requested restriction. The lens asks for a justified and accountable decision, not a presumption that every request must prevail. Recognizing a restriction may require recording that a use is prohibited without collecting the underlying knowledge or disclosing its rationale.

The distinction between recognizing authority and merely extracting feedback is central. An institution could improve its information collection while leaving participants unable to govern resulting uses. Equally, more meetings may add cost without influence. Evidence of participation should therefore concern consequential decisions, resourcing, and remedies as well as attendance and publication counts.

4 The legal and policy baseline

The relevant legal landscape is layered. No single privacy policy, sovereignty label, or launch approval can settle every data relationship. Table 2 identifies selected regimes and the limits of what they establish. It is a baseline for research and implementation planning rather than a project-specific legal determination.

Table 2 Selected regimes and their analytical limits

Regime	Relevant scope	What should not be inferred
UNDRIP and Canada's UN Declaration Act	Indigenous rights; federal measures for consistency and an action plan	A complete operational protocol or a blanket answer to every project consent dispute
PIPEDA	Personal information in covered private-sector activities, subject to the Act's scope and exceptions	General collective authority over all environmental, commercial, or community data
Nova Scotia public-sector privacy legislation	Covered public bodies, information practices, and applicable service-provider obligations	Automatic application of the public-sector regime to every private space supplier
GDPR	Personal data processing within its material and territorial scope	General jurisdiction over every Canadian satellite service or a collective sovereignty framework
Remote Sensing Space Systems Act	Covered remote sensing systems and operators, licensing and security controls	A general public or community right to every satellite product

Regime	Relevant scope	What should not be inferred
Project approval conditions	The approved undertaking and its specified obligations	Proof of compliance, consent, or coverage of every downstream satellite application

Canada's United Nations Declaration on the Rights of Indigenous Peoples Act requires the Government of Canada, in consultation and cooperation with Indigenous Peoples, to take all measures necessary to ensure federal laws are consistent with the Declaration; it separately requires the designated federal minister, in consultation and cooperation with Indigenous Peoples and other federal ministers, to prepare and implement an action plan (Canada, 2021, ss. 5–6). This supplies a significant legal commitment. It does not itself specify every permission, governance body, or data architecture proposed here. Community protocols and project agreements need to be assessed in their own contexts rather than treated as automatically created by the Act.

PIPEDA applies to personal information an organization collects, uses, or discloses in covered commercial activities, and to personal information about its employees or job applicants collected, used, or disclosed in connection with operating a federal work, undertaking, or business, subject to the Act's limits and exceptions (Canada, 2000, ss. 2, 4; Schedule 1). Its accountability, purpose, consent, safeguard, access, and challenge provisions provide useful operational questions. However, environmental observations may matter collectively without being personal information. A dataset's exclusion from one privacy regime does not settle its ethical, contractual, Indigenous governance, or security implications.

For Nova Scotia, timing matters. The province announced that its replacement Freedom of Information and Protection of Privacy Act would come into force on 1 April 2027. That future regime should not be described as already operative on 27 October 2026 (Nova Scotia, 2026). The existing public-sector framework and the Personal Information International Disclosure Protection Act must be considered where applicable. The province's guidance describes PIIDPA requirements and exceptions concerning covered public bodies, municipalities, their service providers, and personal information stored or accessed outside Canada (Nova Scotia Department of Justice, n.d.). A locally hosted private-sector service is therefore neither automatically compliant nor automatically subject to every public-body rule.

The GDPR is a useful comparator because it makes purposes, lawful bases, data minimization, confidentiality, and accountability explicit. Its territorial reach depends on Article 3, including establishment and specified offering or monitoring activities concerning people in the Union; it is not triggered simply because a satellite travels over Europe. Its material focus is personal data processing, with defined limits (European Union, 2016, arts. 2–6). Importing GDPR language would not resolve collective authority over territories or nonpersonal knowledge. The appropriate lesson is disciplined specification of obligations and responsibility, followed by assessment of the relevant local regime.

Canada's Remote Sensing Space Systems Act illustrates a different source of authority. The Act restricts operation of covered systems without a licence, establishes licence conditions, and provides powers concerning interruption and priority access (Canada, 2005, ss. 5, 8, 14–15). These provisions constrain promises of unlimited control or uninterrupted access. They do not establish a general community access entitlement. A proposed data agreement should therefore describe what a participating provider can actually promise, which lawful disclosures or service restrictions may override it, and what notice or explanation can lawfully be given.

Policy creates opportunities without settling authority. Canada's satellite Earth observation strategy commits to collaboration with Indigenous communities and a vision of free, open, accessible data

(Canadian Space Agency [CSA], 2022). The 2026 Research Opportunities in Satellite Earth Observation call includes applications for nontraditional users, including Indigenous communities, while limiting eligible recipients to Canadian universities and post-secondary institutions (CSA, 2026, ss. 2–3.1). This offers a concrete example of an institutional partnership route, not guaranteed funding or proof that a governance-only study would qualify. It also raises a design question: how can community priorities shape a project in which a university is the eligible applicant and grant recipient?

The defensible policy gap is consequently a question of translation and coverage. The reviewed regimes address important but different problems. The selected instruments do not, by themselves, establish how community-defined benefit, governance of nonpersonal data, affordable analytical capability, and remedies across the supply chain would operate in the cases proposed here. This is not evidence of illegality. That uncertainty motivates interviews and agreement-level analysis to identify existing arrangements and assess whether additional mechanisms would be justified.

A current policy opening makes this investigation timely. ISED's October 2026 consultation on legislative and regulatory modernization includes ground stations and data within its scope and considers a new Space Act and coordinated administration. It requests comments by 21 December 2026 (Innovation, Science and Economic Development Canada [ISED], 2026, paras. 1–2, 20–27, 82–87). These are consultation proposals, not enacted powers. The lens developed here could inform questions about safeguards for data activities and the design of coordination: whose authority is recorded, how restrictions and obligations travel between institutions, and what routes exist for correction or challenge. Regulatory coordination would not, by itself, establish the authority, representation, or operating mandate of a community governance service.

5 Four applications to investigate

The following applications are proposed cases, not findings about community demand or existing service quality. Each starts with a decision, identifies the data relationship, and asks what authority and capability would make the information useful. They are connected by the lens but need not share one database or institution.

Launch information and fisheries coordination

The decision problem is whether people using affected waters can plan around authoritative restrictions and changes, communicate relevant concerns, and receive a response. A first study could trace a documented launch information episode, or use an explicitly simulated episode where appropriate, from the responsible authority through the relevant operator and communication channels to intended recipients. It would distinguish the legally authoritative notice from an additional communication service. An unofficial map must not become a competing source of navigational instruction.

A candidate service could carry notice identifiers, current versions, effective times, boundaries, changes, cancellations, and responsible contacts in formats selected with users. The authoritative source, update responsibility, and procedure for conflicting information would have to be agreed before operational use. A website alone may not establish receipt or comprehension; interviews and a tabletop exercise could compare channels and identify where delays arise. Researchers should not prescribe a universal notice period from historical plans or draft guidance.

Coordination may also create requests for information from fishers. The design question is the minimum information needed for the defined purpose. A seasonal, generalized activity layer might sometimes

suffice where individual routes or catch locations would expose sensitive interests. That possibility requires testing: aggregation could remove detail essential to a safe decision. Access to necessary safety notices should not depend on contributing commercially or culturally sensitive data. Potential benefits include less planning uncertainty and a clearer response route; these remain hypotheses requiring a baseline and users' own assessment.

Environmental monitoring and independent interpretation

The decision problem is how a community-defined concern becomes a monitoring question, an observation, an interpretation, and a response. Existing approval conditions offer a place to examine this chain without assuming their processes are deficient. A study should distinguish regulatory monitoring, independent community monitoring, and knowledge supplied under particular protocols. These sources may complement one another while retaining different purposes and authorities.

An implementation option would fund community-selected stewardship and analysis alongside agreed measurement procedures. Records could identify sampling method, calibration, time, location precision, quality limitations, interpretation, and the person responsible for follow-up. Separate disclosure levels could allow a public summary without exposing restricted locations or knowledge. Conflicting interpretations should remain visible with their grounds rather than be silently harmonized into a single official account.

The review by Mamun and Natcher (2023) identifies both the promise of community monitoring and recurring constraints involving funding, training, support, and data use. The review cautions that reported outcomes reflect researchers' accounts rather than the perspectives of participating communities; it does not establish a benefit at this spaceport. It supports treating ongoing stewardship as a cost to fund and evaluate. The Mi'kmaw Conservation Group's undated website describes a technical-resource role for communities, an advisory board of fisheries managers, community rights holders, and Elders, and examples of monitoring activity (Mi'kmaw Conservation Group, n.d.). This records the organization's own description; the page includes older project records and does not verify current staffing, project status, or willingness to participate in a space project. Research should begin by identifying such existing arrangements and asking what, if anything, they would want from satellite or launch-sector partners.

Procurement and workforce participation

The decision problem is whether people and organizations can understand opportunities, meet requirements, and assess how benefits are distributed. A candidate information service could provide procurement categories, qualification requirements, deadlines, relevant training, and named assistance. It should distinguish information useful to prospective suppliers from competitive bids, individual applications, worker concerns, and commercially restricted technical records. A general commitment to Canadian expenditure is not a measure of local or Indigenous procurement.

Researchers would first need explicit definitions and time periods and, for quantitative rates or shares, denominators and appropriate comparisons. Are figures referring to all expenditure, eligible procurement, contract value, paid invoices, jobs, or training places? What counts as local, Indigenous-owned, or a sustained employment outcome, and who authorizes those classifications? Changing definitions midway could create apparent success without a corresponding change in opportunity. Small suppliers should be asked about time, certification costs, cash flow, contract size, and reporting burden as well as information access.

Workers require a distinct route into the research. Employer-selected respondents may not safely describe monitoring, exclusion, or retaliation concerns. Recruitment and reporting should protect confidentiality without promising anonymity that a small sector cannot guarantee. Possible improvements include clearer opportunity information and funded routes to qualification. Possible constraints involving suitable contracts, bargaining power, childcare, transport, or training capacity may require changes beyond an information service. Those are alternative explanations to investigate before describing information infrastructure as the remedy.

Community-directed satellite services

The decision problem is whether a specific satellite product improves a locally selected task, such as interpreting coastal change or supporting marine planning. Begin with the decision and compare existing information sources before choosing a sensor. An Atlantic launch industry may offer partnerships, technical expertise, procurement, or training; the required observation may come from a satellite launched elsewhere. Local launch and community use are separate relationships.

Satellite products require technical and governance validation. Paolo et al. (2024) demonstrate the value of combining imagery and vessel information while documenting limitations of public automatic identification system data. Their global results do not establish the coverage or detection performance of a proposed service for small vessels near Canso. A missing public track is not proof of absence or illegal fishing. A regional application would need appropriate reference observations, coverage assessment, spatial resolution, revisit interval, delivery latency, relevant acquisition limitations, uncertainty, sensor and algorithm versions, and careful interpretation of what can actually be detected.

The governance review must also follow inference. A map made by combining openly available imagery with restricted harvesting information may reveal more than either source alone. A service intended for planning should not silently become an enforcement, insurance, or commercial intelligence product. Any proposed secondary use needs a fresh examination of authority, applicable obligations, risks, and permissions. The ability to refuse a particular linkage may be as significant as access to more data. Benefits should be assessed through validated usefulness for an agreed task and acceptable distribution of costs and exposure, not through image availability alone.

6 How an arrangement could be implemented

Choose an institution after establishing the problem

The first implementation deliverable should be a case-specific governance specification. It would identify the decision, existing institutions, relevant authorities, data flows, applicable constraints, proposed service, accountable owner, resource commitments, evaluation, and exit conditions. It could support a procurement decision or a small trial without presuming a regional hub.

Paprica et al. (2023) provide an operational bridge through requirements for data collaborations covering legal authority, accountable governance, management, data users, and engagement. Their requirements were tested and refined with data-focused organizations and initiatives; this does not establish launch-sector effectiveness. The authors note that their team was concentrated in Canadian and US public and not-for-profit settings, and that commercial application and detailed operational guidance for Indigenous Nations require further work. Their governance discussion supports asking whether a named board has a demonstrable connection to actual data decisions. A governance diagram is insufficient if no one can answer a request, correct a record, or fund an agreed task.

Table 3 compares options to evaluate. These are proposed arrangements. Their legal validity, costs, and legitimacy depend on the parties and use case. Calling an intermediary a data trust does not establish a valid trust, fiduciary duties, or authority to transfer every relevant right. The options can be combined: federation is a technical delivery pattern that can operate under several different governance arrangements.

Table 3 Governance and delivery options for a bounded use case

Option	When it may fit	Main issue to test
Improve existing processes	A notice or response problem within existing mandates	Whether clearer duties and resources solve the problem
Bilateral or small multilateral agreement	A few parties exchanging specified information	Bargaining imbalance and repeated negotiation
Shared stewardship service	Repeated requests needing support and review	Mandate, independence, liability, and sustainable funding
Member cooperative	A willing peer group pooling services	Exclusion of nonmembers and limits of representation
Legally constituted trust	Defined rights and beneficiaries suited to that vehicle	Valid legal form, transferable authority, and enforceable duties
Federated exchange	Several custodians needing repeated bounded access	Coordination costs, security, usability, and downstream enforcement

An operator-funded intermediary could be useful while remaining vulnerable to capture. Its arrangements would need to specify appointment and removal, conflicts, budget continuity, decision authority, review, and the ability to publish authorized adverse findings. A member cooperative may support participating fishers or suppliers, but should not be presumed to represent nonmembers. An Indigenous Nation may choose a service relationship while retaining its own authority, protocols, and custodian. Institutional options should not be selected by majority vote across unlike rights claims.

Specify the data relationship before building interfaces

A practical specification could record the data's source and steward; the authority supporting each proposed use; relevant individual and collective permissions; purposes and prohibited uses; approved recipients and subprocessors; sensitivity and location precision; linkage and inference limits; retention; incident response; mandatory disclosure constraints; and correction or dispute routes. It should identify who can change these terms and which changes require renewed authorization. Some values may remain disputed; recording disagreement is preferable to inventing consensus.

Access should be proportionate to the use case. Some information may be public; some may be shared under conditions; some may remain with its custodian and support only an approved analysis; some should not be collected. Even a catalogue can expose sensitive locations or relationships, so metadata requires review. A common vocabulary must preserve meaningful differences in community categories and uncertainty. Technical interoperability should be evaluated alongside the fidelity of translation and the consequences of any lost context.

Existing standards can support parts of this work. W3C's ODRL information model represents permissions, prohibitions, duties, and constraints, while PROV describes relationships among data, activities, and agents (Iannella & Villata, 2018; Groth & Moreau, 2013). Neither determines legitimate authority or proves compliance. Community-chosen Local Contexts labels can communicate knowledge protocols and provenance, but the organization describes them as extra-legal tools complementing existing legal arrangements (Local Contexts, 2026). These are optional components, not substitutes for negotiated authority or actual controls.

A federated design could leave records with existing custodians while providing authorized discovery or exchange. It could also impose expensive interfaces, fragmented security responsibilities, and a common schema shaped by the strongest participant. The OECD's October 2026 analysis of data spaces identifies coordination and viability problems and makes a particularly important enforcement distinction: after transfer, usage control depends partly on recipient compliance; contracts, provenance, and audit supplement technical controls (OECD, 2026, ch. 3). A system should not promise recall of every downloaded copy or perfect enforcement of purpose restrictions.

Resource authority and make feedback consequential

The resource plan should include paid stewardship, participation time, independent interpretation, accessible communication, training, maintenance, security, and dispute resolution. It should distinguish one-time development from recurring operation. Funding should follow responsibilities: a community asked to review repeated requests needs capacity to do so. A notional right that is too costly to exercise should be recorded as an implementation problem, not counted as achieved sovereignty.

Policy can enter through several instruments: a community's own protocols, project agreements, procurement terms, grant conditions, organizational policies, and regulatory requirements where the responsible authority has power to impose them. The paper does not assume every regulator can add any desired condition. Each proposal needs an actor with the relevant authority, a lawful instrument, an accountable operator, and a funded process. Public funders and buyers may be able to support stewardship or require specified reporting within their mandates; the applicable agreement must be examined before making that claim for a particular project.

Implementation would then proceed by gates rather than a fixed schedule. First, establish a recognized need and the proper authorities. Second, document the current process and baseline. Third, agree on a narrow arrangement, resources, and limits. Fourth, test it with scenarios and nonsensitive or explicitly authorized material. Fifth, evaluate whether to operate, revise, expand, or stop it. This sequence is a design proposal, not evidence of a community mandate. It should allow a finding that improved communication, an existing institution, or no new system is the appropriate outcome.

7 Research needed to determine whether it would work

Governance and ethics before recruitment

The next phase should be a comparative qualitative study of a small number of data relationships, with interviews supported by document and process analysis. Research involving Mi'kmaw communities, knowledge, lands, or rights needs an appropriate governance process established with the relevant authorities. TCPS 2 Chapter 9 addresses engagement, community agreements, diversity of authority, individual consent, benefits, and interpretation (Canadian Institutes of Health Research et al., 2022, arts.

9.1–9.3, 9.5–9.7, 9.11, 9.13–9.17). It should not be reduced to either an individual consent form or an assumption that one leader can authorize every disclosure.

Current Mi'kmaw Ethics Watch guidance should be consulted alongside institutional research ethics requirements and the processes identified by the relevant communities. Its current application page says research without direct participants may still require review and directs researchers who are unsure about an exemption to seek MEW's assessment (Mi'kmaw Ethics Watch, n.d.). Review requirements for this desk review and any subsequent empirical phase should be confirmed with MEW and the relevant institution; public availability should not be treated as an automatic exemption. Required approvals for interviews or access to protected material should be obtained before those activities begin. This paper claims no community or ethics-body endorsement of its lens. Preliminary relationship-building should be distinguished from research data collection; plans to use exploratory remarks should be disclosed in the research proposal with appropriate consent procedures (Canadian Institutes of Health Research et al., 2022, ch. 10, art. 10.1).

Agreements should address recording, storage, access, attribution, interpretation, publication review, future use, return or disposal, and the practical limits of withdrawal. Researchers must distinguish review of sensitive material and authorized collective knowledge from an unrestricted promise that any institutional participant can suppress criticism. Consent should explain the actual risks of recognition in a small sector, lawful disclosure limits, and how disagreements will be reported. Restricted cultural, operational, or commercial information should not be solicited merely to make the analysis more detailed.

Who can answer which questions

The study should seek evidence from several organizations and affected groups. Table 4 identifies evidence holders rather than a confirmed recruitment list. Participation and representation would need to be negotiated. Interviews with organizations should be paired with the experiences of people using, supplying, or being affected by the information.

Table 4 Questions and evidence holders

Research question	People or institutions to approach	Additional evidence
Whose authority applies and what should remain restricted?	Mi'kmaw authorities and knowledge governance roles identified through their own processes	Authorized protocols, mandates, agreements
What information is useful for marine coordination?	Independent fishers, fishing organizations, marine authorities, operator safety staff	Notice histories, version changes, scenario tests
How are monitoring concerns interpreted and answered?	Community-selected stewards, environmental regulator, monitoring specialists, approval holder	Plans, sampling metadata, responses, complaints
What controls access and secondary use?	Satellite and ground-service operators, customers, processors, privacy and security staff	Licences, contracts, data-flow maps, access rules

Research question	People or institutions to approach	Additional evidence
Who can participate economically and at what cost?	Small suppliers, workers and representatives, training providers, buyers, public funders	Requirements, procurement definitions, time and cost records
Who can change a decision or obtain a remedy?	Responsible decision makers and affected participants separately	Terms of reference, reasons, corrections, escalation records

An initial planning envelope of approximately twenty interviews could support scoping, but is not a scientifically guaranteed sample size. Final recruitment should be justified by the study aims, sample specificity, theoretical framing, quality of dialogue, and analysis strategy, rather than a predetermined count (Malterud et al., 2016). It may require more interviews, fewer cases, or a separate phase. Neither twenty interviews nor the use of thematic coding establishes saturation, representativeness, consensus, or an estimate of causal effects. Missing perspectives and reasons for nonparticipation should be reported when doing so is safe and appropriate.

Collect evidence about decisions rather than aspirations alone

Questions should begin with concrete episodes. What information was requested? Who decided whether it could be shared? Which rule was used? Could the affected party check the interpretation? What happened when a concern was raised? What alternative would the respondent prefer? Where participants authorize access, an interview account should be compared with a notice, agreement, dictionary, complaint response, or permission record. Researchers should record whether the evidence shows intended policy, routine practice, an exception, or a hypothetical preference.

A case file could pair a data-flow map with an authority map and a decision history. Qualitative coding would begin with the seven dimensions in Table 1 while allowing new themes and challenges to the framework. Differences between sources should remain visible. Respondent review could check accuracy and agreed confidentiality; it would not certify truth or compel consensus. Comparison across cases could show whether a missing mandate, unclear terminology, technical limitation, funding constraint, or disagreement about the project explains the problem better than missing data infrastructure.

The study should compare candidate arrangements with the current process and an improvement requiring no new platform. Outputs would include authorized case descriptions, a map of unresolved questions, a revised lens, and one or more governance specifications where participants want them. A subsequent pilot would be a separate implementation stage requiring its own authority, resources, evaluation, and controls. The proposed interviews could identify requirements and constraints, but would not by themselves validate a prototype's safety, technical performance, or net economic effects.

8 Evaluation counterarguments and limits

The lens should be capable of producing inconvenient findings. Existing institutions may already provide what participants need. A data-sharing request may be inappropriate. Community members may decline participation. An arrangement may be unaffordable, or a satellite product may add little beyond an existing observation. These outcomes would refine the research argument rather than count as failures to adopt a preferred system.

Three questions can organize later evaluation. First, when a decision role is linked to a named responsible actor, an applicable instrument, and funded support, does the relevant party gain a usable ability to condition a use, correct a representation, or obtain a remedy? Cases in which formalization changes nothing, or informal arrangements work better, would challenge the proposed mechanism. Second, can a specified service achieve its agreed quality or safety requirements while transferring less sensitive information than the current process, and at an acceptable cost? The answer could favour a more complete exchange or no new service. Third, does a satellite or information service improve an agreed decision compared with existing sources and a feasible alternative requiring no new platform? A lack of improvement, or evidence that jurisdiction, finance, or mistrust is the binding constraint, would weigh against the proposed intervention. These questions guide process tracing and later evaluation; they are not causal findings of this review.

Evaluation should separate procedural performance, substantive authority, service quality, and distribution. Procedural measures could include response times and completeness of reasons. Authority could be examined through a real or simulated request to condition a use, obtain an approved analysis, correct a representation, or escalate a dispute. Service quality would depend on the case: notice comprehension, measurement reliability, or product usefulness. Distribution would examine whose time and money are consumed and whose priorities are served. Counts of meetings, datasets, downloads, or contracts are contextual indicators rather than sufficient measures of sovereignty or equitable benefit.

Targets should be set after baseline measurement and with the relevant participants. Before-and-after changes should be interpreted cautiously because launches, funding, personnel, weather, or market conditions may change simultaneously. A small qualitative study can identify plausible mechanisms and constraints. Demonstrating net economic impact, reduced environmental harm, or durable improvements in decision power would require additional data and an appropriate evaluation design. No quantitative regional benefit estimate follows from this review.

There are substantive tensions. Public environmental accountability may require disclosure that a company would prefer to restrict. Individual privacy and collective access may point in different directions. A safety requirement or lawful state order may limit a negotiated permission. Open science may support useful analysis while creating risks for restricted knowledge. The response should be an explicit process for identifying authority, necessity, proportionality, lawful limits, and remedy. A technical interface cannot adjudicate these conflicts by itself.

The evidence base also limits the paper. Public documents do not establish the adequacy of current arrangements, and provider announcements are not independent performance evaluations. Principles and guidance establish important normative and operational questions, while studies from health data collaborations, global maritime observation, and Earth systems science require contextual transfer. The proposed framework has not been tested with Atlantic participants. The review therefore supports a focused investigation and conditional design options, not a claim of local endorsement, universal novelty, noncompliance, or proven benefits.

9 Conclusion

Atlantic Canadian space infrastructure could support data sovereignty through the decisions and institutions around it: who defines monitoring questions, authorizes disclosure, interprets observations, controls secondary use, receives usable information, and can obtain a remedy. Physical location, national ownership, cybersecurity, and consultation can contribute to parts of that arrangement. None establishes the whole relationship.

The strongest research contribution is to make those relationships inspectable while preserving the distinct foundation of Indigenous data sovereignty. A practical next study would trace selected data relationships with appropriately authorized Mi'kmaw participation and separately recruited community, industry, worker, and government perspectives. It would test existing arrangements and alternatives before recommending infrastructure. Where a change is justified, implementation requires a recognized authority, an appropriate instrument, funded capability, defined permissions, and reviewable consequences. This provides a pathway for investigating how a governance principle becomes policy and practice and how its outcomes could be assessed.

References

Canada. (2000). Personal Information Protection and Electronic Documents Act, S.C. 2000, c. 5. https://www.laws.justice.gc.ca/eng/acts/P-8.6/FullText.html

Canada. (2005). Remote Sensing Space Systems Act, S.C. 2005, c. 45. https://lois-laws.justice.gc.ca/eng/acts/R-5.4/FullText.html

Canada. (2021). United Nations Declaration on the Rights of Indigenous Peoples Act, S.C. 2021, c. 14. https://laws.justice.gc.ca/eng/acts/U-2.2/FullText.html

Canadian Institutes of Health Research, Natural Sciences and Engineering Research Council of Canada, & Social Sciences and Humanities Research Council of Canada. (2022). Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans [Chapters 9 and 10]. https://ethique.gc.ca/eng/tcps2-eptc2_2022_chapter9-chapitre9.html https://ethics.gc.ca/eng/tcps2-eptc2_2022_chapter10-chapitre10.html

Canadian Space Agency. (2022, January 20). Resourceful, resilient, ready: Canada's Strategy for Satellite Earth Observation. https://www.asc-csa.gc.ca/eng/publications/canada-strategy-for-satellite-earth-observation/05-about-this-strategy-a-bold-leap-into-the-future.asp

Canadian Space Agency. (2026, August 31). Research Opportunities in Satellite Earth Observation [Announcement of opportunity]. https://www.asc-csa.gc.ca/eng/funding-programs/funding-opportunities/ao/2026-research-opportunities-in-satellite-earth-observation.asp

Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., & Hudson, M. (2020). The CARE Principles for Indigenous Data Governance. Data Science Journal, 19, Article 43. https://doi.org/10.5334/dsj-2020-043

Department of National Defence. (2026, March 16). Minister McGuinty announces strategic investments in sovereign space launch [News release]. https://www.canada.ca/en/department-national-defence/news/2026/03/minister-mcguinty-announces-strategic-investments-in-sovereign-space-launch.html

European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 [General Data Protection Regulation]. Official Journal of the European Union, L 119, 1–88. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679

First Nations Information Governance Centre. (n.d.). The First Nations Principles of OCAP®. Retrieved October 27, 2026. https://fnigc.ca/ocap-training/

Groth, P., & Moreau, L. (Eds.). (2013, April 30). PROV-Overview: An overview of the PROV family of documents [W3C Working Group Note]. World Wide Web Consortium. https://www.w3.org/TR/prov-overview/

Hummel, P., Braun, M., Tretter, M., & Dabrock, P. (2021). Data sovereignty: A review. Big Data & Society, 8(1). https://doi.org/10.1177/2053951720982012

Iannella, R., & Villata, S. (Eds.). (2018, February 15). ODRL Information Model 2.2 [W3C Recommendation]. World Wide Web Consortium. https://www.w3.org/TR/odrl-model/

Innovation, Science and Economic Development Canada. (2026, October). Consultation on legislative and regulatory modernization for space activities (SPB-007-26). https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en/learn-more/key-documents/consultations/consultation-legislative-and-regulatory-modernization-space-activities

Jennings, L., Jones, K., Taitingfong, R., Martinez, A., David-Chavez, D., Alegado, R. ‘A., Tofighi-Niaki, A., Maldonado, J., Thomas, B., Dye, D., Weber, J., Spellman, K. V., Ketchum, S., Duerr, R., Johnson, N., Balch, J., & Carroll, S. R. (2025). Governance of Indigenous data in open earth systems science. Nature Communications, 16, Article 572. https://doi.org/10.1038/s41467-024-53480-2

Leaf Space. (2026, September 8). Leaf Space extends its global network with first ground station in Canada's Spaceport Nova Scotia. https://leaf.space/leaf-space-extends-its-global-network-with-first-ground-station-in-canadas-spaceport-nova-scotia-2/

Local Contexts. (2026, September 10). Frequently asked questions. https://localcontexts.org/support/frequently-asked-questions/

Malterud, K., Siersma, V. D., & Guassora, A. D. (2016). Sample size in qualitative interview studies: Guided by information power. Qualitative Health Research, 26(13), 1753–1760. https://doi.org/10.1177/1049732315617444

Mamun, A. A., & Natcher, D. C. (2023). The promise and pitfalls of community-based monitoring with a focus on Canadian examples. Environmental Monitoring and Assessment, 195, Article 445. https://doi.org/10.1007/s10661-022-10841-y

Maritime Launch Services. (2026, September 3). Join our community liaison committee. https://www.maritimelaunch.com/news/join-community-liaison-committee-2026

Mi'kmaw Conservation Group. (n.d.). Current projects. Retrieved October 27, 2026. https://mikmawconservation.ca/programs-and-services/current-projects/

Mi'kmaw Ethics Watch. (n.d.). Apply for Mi'kmaw research approval. Mawkim. Retrieved October 27, 2026. https://mawkim.org/mikmaw-ethics-watch/apply-for-research-approval/

Nova Scotia. (2026, October 1). Nova Scotia among Canada's leaders in FOIPOP response rates [News release]. Service Nova Scotia. https://news.novascotia.ca/en/2026/10/01/nova-scotia-among-canadas-leaders-foipop-response-rates

Nova Scotia Department of Justice. (n.d.). FAQ: Personal Information International Disclosure Protection Act. Retrieved October 27, 2026. https://novascotia.ca/just/iap/piidpaquest.asp

Nova Scotia Environment. (n.d.). Canso Spaceport Facility Project [Continuing environmental assessment register; ministerial decision dated June 4, 2019]. Retrieved October 27, 2026. https://novascotia.ca/nse/ea/canso-spaceport-facility/

Nova Scotia Environment. (2019, June 4). Environmental assessment approval: Canso Spaceport Facility Project, Maritime Launch Services Limited—Terms and conditions for environmental assessment approval. https://novascotia.ca/nse/ea/canso-spaceport-facility/Conditions.pdf

Organisation for Economic Co-operation and Development. (2026, October 20). Data spaces between co-ordination challenges, financial viability and public policy. OECD Publishing. https://doi.org/10.1787/0066f11b-en

Paolo, F. S., Kroodsma, D., Raynor, J., Hochberg, T., Davis, P., Cleary, J., Marsaglia, L., Orofino, S., Thomas, C., & Halpin, P. (2024). Satellite mapping reveals extensive industrial activity at sea. Nature, 625, 85–91. https://doi.org/10.1038/s41586-023-06825-8

Paprica, P. A., Crichlow, M., Curtis Maillet, D., Kesselring, S., Pow, C., Scarnecchia, T. P., Schull, M. J., Cartagena, R. G., Cumyn, A., Dostmohammad, S., Elliston, K. O., Greiver, M., Hawn Nelson, A., Hill, S. L., Isaranuwatchai, W., Loukipoudis, E., McDonald, J. T., McLaughlin, J. R., Rabinowitz, A., … McGrail, K. (2023). Essential requirements for the governance and management of data trusts, data repositories, and other data collaborations. International Journal of Population Data Science, 8(4), Article 2142. https://doi.org/10.23889/ijpds.v8i4.2142

Strum Consulting. (2018, June). Canso Spaceport Facility Environmental Assessment Registration Document. Prepared for Maritime Launch Services Limited. https://novascotia.ca/nse/ea/canso-spaceport-facility/registration_document.pdf

Transport Canada. (n.d.). Application requirements for obtaining authorization to launch uncrewed orbital and suborbital rockets in Canada [Draft; version table through 0.4, April 30, 2025]. Retrieved October 27, 2026. https://tc.canada.ca/en/aviation/commercial-space-launches-canada/application-requirements-obtaining-authorization-launch-uncrewed-orbital-suborbital-rockets-canada

Transport Canada. (2026a, April 23). Commercial space launches in Canada [Page modified]. https://tc.canada.ca/en/aviation/commercial-space-launches-canada

Transport Canada. (2026b, March 20). Consultation on commercial space launch in Canada [Page modified]. https://tc.canada.ca/en/corporate-services/consultations/consultation-commercial-space-launch-canada

United Nations. (2007). United Nations Declaration on the Rights of Indigenous Peoples (A/RES/61/295). https://laws.justice.gc.ca/eng/acts/U-2.2/FullText.html